

METHODS: MODELED EXTENT AND BOUNDARY STATISTICS

Modeled extent

The modeled extent is a rectangular bounding box enclosing the USGS GAP range polygon for the species after that polygon has been buffered outward by a fixed 250 km margin, computed in a projected equal-area coordinate system (EPSG:5070, USA Contiguous Albers Equal Area Conic). The buffer is applied uniformly across species rather than sized to each species individually. This keeps the boundary condition standardized across the framework and avoids using a species' own centroid velocity, which is an output of the analysis, to set a parameter of the analysis.

Derivation of the 250 km buffer

The 250 km figure derives from Huang, Sauer, and Dubayah (2017), who tracked the abundance-weighted geographic centroid of 57 permanent resident North American bird species over 44 years using Breeding Bird Survey data and found an average, multidirectional centroid velocity of 5.89 km/yr. Projected across rENM's 45-year study window this gives approximately 265 km, close to the 259 km average displacement that study reports directly over a comparable period. A slightly conservative round figure of 250 km is used.

This full, multidirectional centroid velocity was chosen over other available bird range-shift estimates that report only a poleward or latitudinal component of movement. rENM computes a full two-dimensional suitability centroid, and the pilot species examined during development did not shift uniformly poleward.

What the buffer does and does not affect

All range-based summary statistics — positive and negative trend percentage, hot spot area and percentage — are computed against the unbuffered GAP range polygon, not against the bounding box. The buffer and its enclosing rectangle exist only to keep the trend surface from being artificially clipped at the historic range edge; they do not redefine what counts as range in the reported tables.

Trend behavior within the buffer ring itself — the zone lying between the raw GAP polygon and the 250 km buffered polygon — is reported separately as a boundary trend statistic. Comparing the ring against the range interior indicates whether conditions just outside the historic range are improving or deteriorating relative to the range the species occupies, which a static range polygon cannot show.

Limitations

The 250 km figure is drawn from permanent resident species specifically, chosen in the source study because a resident species' abundance centroid is a cleaner climate signal than a migrant's, whose breeding and wintering ranges can shift somewhat independently. It may therefore understate or misrepresent the appropriate margin for strongly migratory species. This is a known limitation rather than a resolved one.

The buffer distance is a provisional, literature-informed default. It should be revisited as more species run through the pipeline and as taxon- or guild-specific velocity estimates become practical to compute internally.

Reference

Huang, Q., J. R. Sauer, and R. O. Dubayah. 2017. Multidirectional abundance shifts among North American birds and the relative influence of multifaceted climate factors. *Global Change Biology* 23:3610–3622. <https://doi.org/10.1111/gcb.13683>